

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – 26047

Problem Statement Title – Patient Case-Taking Software.

Theme – MedTech / BioTech / HealthTech

PS Category – Software

Team Name – SwasthyaSetu





EXISTING PROBLEMS

1



Static & Lengthy Case Taking:

Patients answer fixed questions even when symptoms differ.

2



Fragmented Patient Records:

Previous prescriptions, reports and consultations are difficult to connect.

3



Complex Ayurvedic Case Taking:

Detailed Ayurvedic parameters require additional time and effort.

4



Language Barriers:

Patients may struggle to describe symptoms or understand medical forms.

5



Manual Case Summarization:

Doctors spend valuable consultation time reviewing and summarizing information.



OUR SOLUTION – MEDIKIOSK



Adaptive Case Taking:

AI changes questions according to the patient's symptoms and answers.



Unified Patient Record:

Symptoms, history, reports and previous consultations are connected.



Integrated Ayurveda:

Ayurvedic parameters are incorporated into the digital case record.



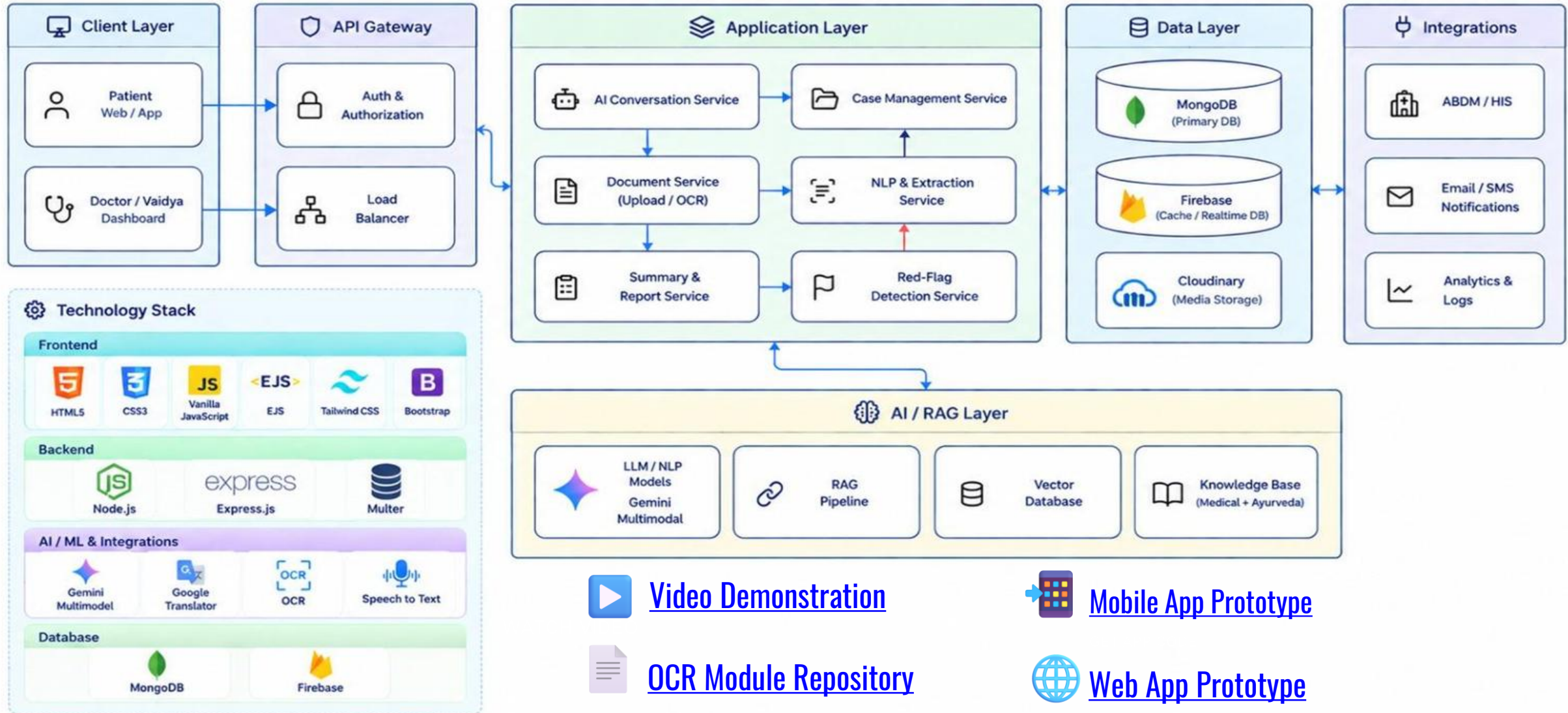
Multilingual Interaction:

Patients can communicate through voice/touch in their preferred language.



AI Case Summarization:

Long conversations and records become a concise structured clinical summary.





IMPLEMENTATION FEASIBILITY



BUILT ON EXISTING SYSTEMS

Leverages patient management, medical records and clinical workflows.



MODULAR DEPLOYMENT

Can be implemented in phases with minimal changes.



COST-EFFECTIVE SETUP

Uses existing infrastructure and open technologies.



SEAMLESS INTEGRATION

FHIR R4 / ABDM aligned interoperability.



SCALABLE & FUTURE-READY

Cloud-native architecture for easy expansion.



VIABILITY & RISK MITIGATION



HANDWRITTEN DOCUMENTS (OCR)

Risk: Variable OCR accuracy

Mitigation: AI confidence scoring + doctor verification.



REGIONAL ACCENTS

Risk: Multi-dialect input variation

Mitigation: Multilingual ASR + touch-based fallback.



AI HALLUCINATION

Risk: Inaccurate information

Mitigation: Restricted output + mandatory doctor review.



DATA PRIVACY

Risk: Sensitive patient data

Mitigation: ABDM compliance + end-to-end encrypted storage.



BUSINESS POTENTIAL



IMPROVED PATIENT ACCESS

Multilingual and voice-based intake for wider adoption.



OPERATIONAL COST REDUCTION

Automates data collection and reduces manual effort.



NEW REVENUE OPPORTUNITIES

Integration with digital health programs and public healthcare systems.



STRONGER HEALTHCARE ECOSYSTEM

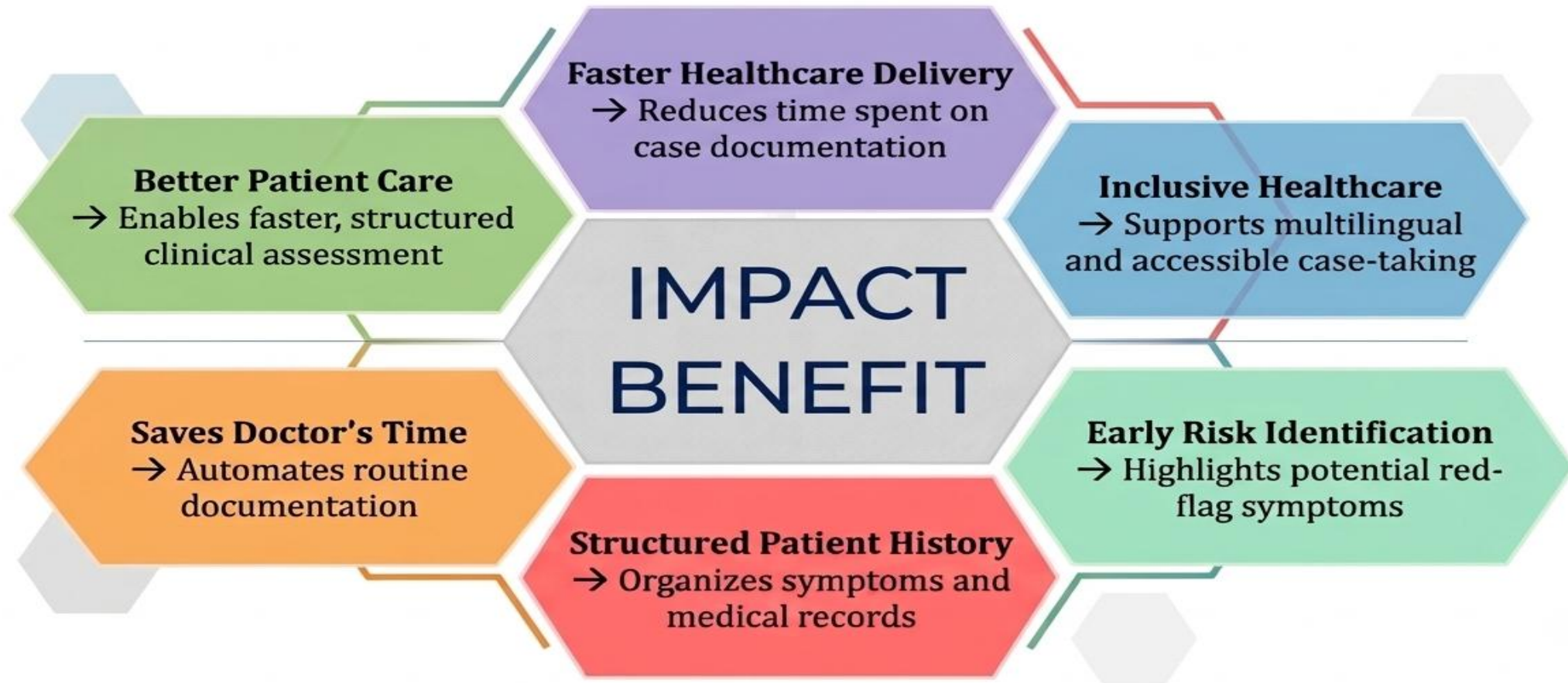
Supports government initiatives like ABDM and Ayushman Bharat.



AI ASSISTS. DOCTOR DECIDES.

Augments clinical workflow while keeping a human in control.

IMPACT AND BENEFITS





PATIENT HISTORY & AI DIAGNOSIS

Uses patient history and clinical data to support diagnosis and clinical decisions.

Source: <https://ieeexplore.ieee.org/document/9643300>



INDIAN LANGUAGE CLINICAL ASR

Voice-based patient input can be processed across regional languages to improve accessibility.

Source: <https://ieeexplore.ieee.org/abstract/document/9676651>



FHIR INTEROPERABILITY

Enables standardized and seamless exchange of patient health data between healthcare systems.

Source: <https://ieeexplore.ieee.org/document/6627810>



PATIENT INTAKE KIOSKS

Self-service kiosk for collecting patient information and reducing waiting time.

Source: <https://ieeexplore.ieee.org/document/10253807>



MEDICAL RECORD DIGITIZATION(OCR)

OCR digitizes medical records into structured, machine-readable patient data.

Source: <https://ieeexplore.ieee.org/document/10222038>



DIGITAL HEALTH IN INDIA

Digital technologies can improve healthcare accessibility, efficiency, and delivery across India.

Source: <https://ieeexplore.ieee.org/document/9441465>